

ReactJS HOL 9

ES6 Features in React

In this exercise, I developed a React application named **cricketapp** to understand the ES6 features used in React. I implemented the `map()` method to display a list of cricket players and used arrow functions to filter players with scores less than or equal to 70. I also applied array destructuring to display odd and even team players separately and used the spread operator to merge two arrays of players. The application was executed successfully and produced the expected output.

Source Code

ListOfPlayers.js

```
import React from "react";

export function ListOfPlayers(props) {
  return (
    <div>
      <ul>
        {
          props.players.map((item) => {
            return (
              <div key={item.name}>
                <li>
                  Mr. {item.name}
                  <span> {item.score}</span>
                </li>
              </div>
            );
          })
        }
      </ul>
    </div>
  );
}
```

```

        }
      </ul>
    </div>
  );
}

export function Scorebelow70(props) {
  let players70 = [];
  props.players.map((item) => {
    if (item.score <= 70) {
      players70.push(item);
    }
  });
  return null;
});

return (
  <div>
    <ul>
      {
        players70.map((item) => {
          return (
            <div key={item.name}>
              <li>
                Mr. {item.name}
                <span> {item.score}</span>
              </li>
            </div>
          );
        })
      }
    </ul>
  </div>
);

```

```
        })
      }
    </ul>
  </div>
);
}
```

IndianPlayers.js

```
import React from "react";

export function OddPlayers([first, , third, , fifth]) {
  return (
    <div>
      <ul>
        <li>First : {first}</li>
        <li>Third : {third}</li>
        <li>Fifth : {fifth}</li>
      </ul>
    </div>
  );
}

export function EvenPlayers([, second, , fourth, , sixth]) {
  return (
    <div>
      <ul>
        <li>Second : {second}</li>
        <li>Fourth : {fourth}</li>
        <li>Sixth : {sixth}</li>
      </ul>
    </div>
  );
}
```

```
        </ul>
      </div>
    );
  }
const T20Players = [
  "First Player",
  "Second Player",
  "Third Player"
];
const RanjiTrophyPlayers = [
  "Fourth Player",
  "Fifth Player",
  "Sixth Player"
];
export const IndianPlayers = [
  ...T20Players,
  ...RanjiTrophyPlayers
];
export function ListofIndianPlayers(props) {
  return (
    <div>
      <ul>
        {
          props.IndianPlayers.map((player, index) => (
            <li key={index}>Mr. {player}</li>
          ))
        }
      </ul>
    </div>
  );
}
```

```
    }  
  </ul>  
</div>  
);  
}
```

App.js

```
import './App.css';  
  
import { ListOfPlayers, Scorebelow70 } from './ListOfPlayers';  
  
import {  
  OddPlayers,  
  EvenPlayers,  
  IndianPlayers,  
  ListofIndianPlayers,  
} from './IndianPlayers';  
  
function App() {  
  const players = [  
    { name: "Jack", score: 50 },  
    { name: "Michael", score: 70 },  
    { name: "John", score: 40 },  
    { name: "Ann", score: 61 },  
    { name: "Elisabeth", score: 61 },  
    { name: "Sachin", score: 95 },  
    { name: "Dhoni", score: 100 },  
    { name: "Virat", score: 84 },  
    { name: "Jadeja", score: 64 },  
    { name: "Raina", score: 75 },
```

```
{ name: "Rohit", score: 80 }  
];  
const IndianTeam = [  
  "Sachin1",  
  "Dhoni2",  
  "Virat3",  
  "Rohit4",  
  "Yuvaraj5",  
  "Raina6"  
];  
var flag = true;  
if (flag === true) {  
  return (  
    <div>  
      <h1>List of Players</h1>  
      <ListOfPlayers players={players} />  
      <hr />  
      <h1>List of Players having Scores Less than 70</h1>  
      <Scorebelow70 players={players} />  
    </div>  
  );  
} else {  
  return (  
    <div>  
      <h1>Indian Team</h1>
```

```

    <h1>Odd Players</h1>

    {OddPlayers(IndianTeam)}

    <hr />

    <h1>Even Players</h1>

    {EvenPlayers(IndianTeam)}

    <hr />

    <h1>List of Indian Players Merged:</h1>

    <ListOfIndianPlayers IndianPlayers={IndianPlayers} />

</div>

);
}
}

export default App;

```

Figure 1: Project Structure

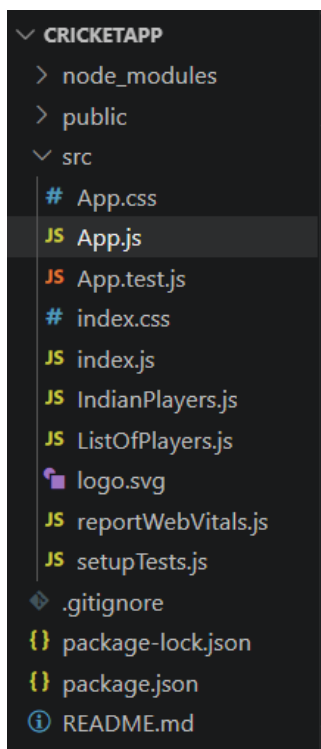


Figure 2: Output when flag = true

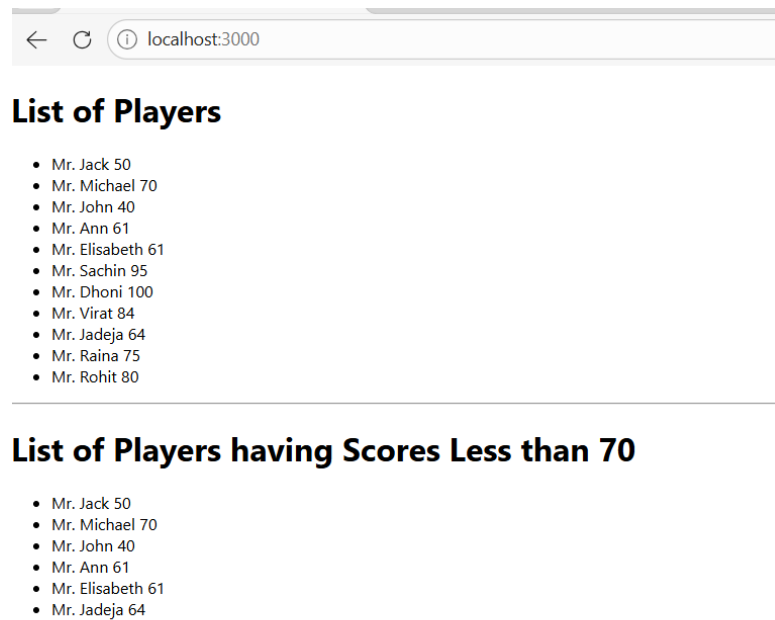


Figure 3: Output when flag = false

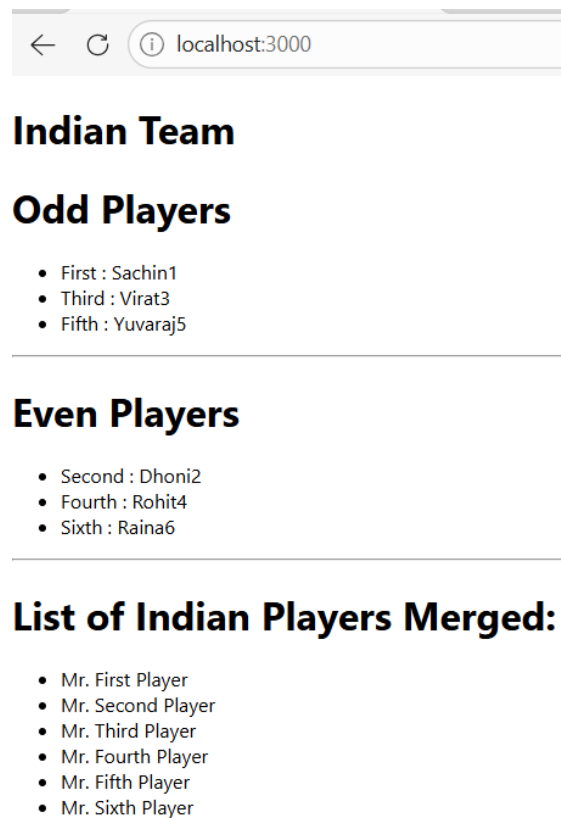
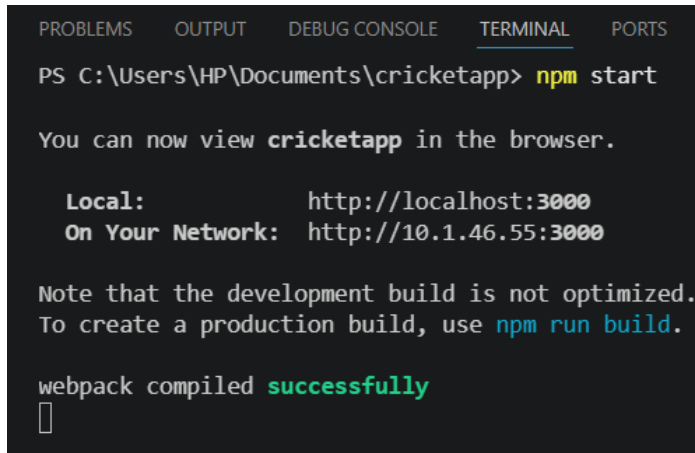


Figure 4: Terminal Output



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\HP\Documents\cricketapp> npm start

You can now view cricketapp in the browser.

  Local:            http://localhost:3000
  On Your Network:  http://10.1.46.55:3000

Note that the development build is not optimized.
To create a production build, use npm run build.

webpack compiled successfully
█
```

Result

The React application was implemented successfully using ES6 features such as `map()`, arrow functions, array destructuring, and the spread operator. The application displayed the list of players, filtered players based on scores, separated odd and even team players, and merged two player arrays successfully. The expected output was obtained.